

How Commodity Indices Are Priced

BCOM and S&P GSCI — roll schedules, the daily roll window, annual rebalancing and market disruption events

A concept guide written to be re-read. Every worked example is computed from the `indexpricing` engine in this repository on **mock** market data, so you can re-run the numbers and change them yourself.

Contract schedules, caps and dates are **illustrative**. They follow the structure of the published methodologies. Always check the current Bloomberg and S&P DJI rulebooks before relying on a detail at work.

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1. The big picture

A commodity index is **not** a basket of physical commodities. It is a set of rules for holding futures contracts. On any day it holds a fixed *quantity* of certain contracts, for example 0.14 barrels of the May WTI contract for every index point. The index level is the value of those holdings, chained from day to day.

Every futures contract expires. So the rules have to say four things:

Question	Rule that answers it	Section
How much of each commodity?	Weighting method: BCOM target % (CIPs converted to CIMs); S&P GSCI production quantities (CPWs)	4
Which contract for each commodity this month?	The designated-contract (roll) schedule, different for each commodity	3.1–3.4
When and how fast to move to the next contract?	The roll window: 20% a day over 5 business days (BCOM days 6–10, GSCI days 5–9)	3.5–3.8
When to reset the quantities?	Annual rebalance, carried out inside the January roll	5
What if a market is broken that day?	Market disruption rules: postpone that commodity's roll or rebalance	6

Three index levels come out of the same holdings:

- **Excess Return (ER):** the headline index (BCOM, SPGSCI ER). It captures futures price changes only, including the gain or loss from rolling.
- **Total Return (TR):** ER plus the interest earned on T-bills held as collateral. This is what a fully funded investor earns.
- **Spot:** the price level of the held contracts. It deliberately ignores the roll, so ER – Spot measures the *roll yield*.

$$\begin{aligned}
 \text{CDR}(t) &= \sum q(c, t-1) \cdot P(c, t) / \sum q(c, t-1) \cdot P(c, t-1) - 1 && \text{contract daily return} \\
 \text{ER}(t) &= \text{ER}(t-1) \cdot (1 + \text{CDR}(t)) \\
 \text{TR}(t) &= \text{TR}(t-1) \cdot (1 + \text{CDR}(t) + \text{TBR}(t)) \\
 \text{TBR}(t) &= [1 / (1 - 91/360 \cdot \text{TB}(t-1))]^{(\text{days}/91)} - 1 && \text{T-bill return} \\
 \text{Spot}(t) &= \sum q(c, t) \cdot P(c, t) / \text{divisor}
 \end{aligned}$$

Here $q(c, t)$ is the quantity of contract c held at the **close** of day t , and P is the settlement price. The key detail is that CDR uses **yesterday's** holdings valued at today's and yesterday's prices. Today's roll trades happen at today's close and only start earning tomorrow. This single detail explains why rolling never makes the ER jump.

2. Futures refresher: why an index must roll

2.1 Month codes

Code	F	G	H	J	K	M	N	Q	U	V	X	Z
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

A ticker such as `CLK22` is root (CL = WTI crude) + month code (K = May) + year (22). Each exchange lists only certain months for each commodity. Crude lists all 12, corn only H K N U Z, and gold is actively traded in G J M Q V Z. That difference is the root of why roll schedules differ from commodity to commodity.

2.2 Expiry, first notice and delivery

Most index commodities are **physically delivered**. A long position still open after **First Notice Day** (FND) can be assigned a delivery. An index never wants barrels of oil or bushels of corn, so it must be completely out of a contract well before FND or Last Trading Day. It therefore rolls in the month *before* the contract's notice period, and often a month or more earlier. It also leaves before the front contract's liquidity dries up.

2.3 Contango, backwardation and roll yield

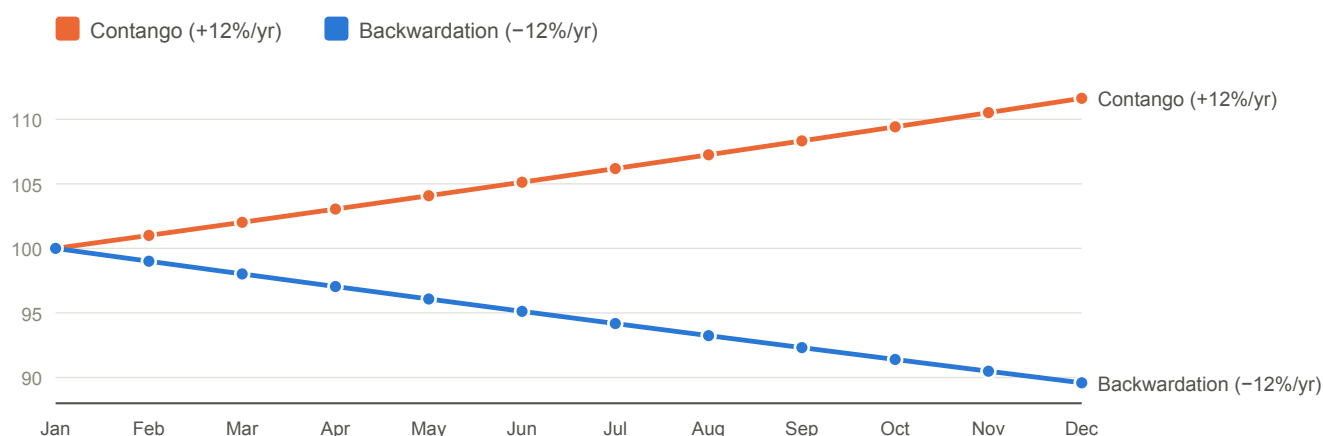


Figure 1. Two illustrative futures curves with the same spot price (100). The x-axis is the contract delivery month.

When the index rolls it sells the expiring contract and buys a later one. In **contango** the later contract is dearer, so each dollar buys fewer barrels. As the curve rolls down toward spot, that position loses value. This is **negative roll yield**. In **backwardation** the later contract is cheaper, and the investor earns **positive roll yield**. Natural gas is the classic steep-contango market, and crude is often backwardated when supply is tight.

Rule of thumb. $ER \approx \text{Spot price change} + \text{roll yield}$. $TR \approx ER + \text{T-bill rate}$. The roll yield is decided almost entirely by the roll schedule and the curve shape at the roll dates. That is why the schedule matters so much.

3. Roll schedules

3.1 The designated contract, and the lead/next pair

For every commodity the index publishes a 12-month table. Each month has a **designated contract**, the contract the index holds once that month's roll is complete. In the code it is a 12-letter string, one month code per calendar month:

```
GC (gold, BCOM):  J J M M Q Q Z Z Z Z G G
                  Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
```

Read it like this: at the end of January the index holds April gold (J). At the end of February it still holds April. In March it moves to June (M), and so on. Within any month two contracts matter:

- **Lead contract** ("current", "old", "front"): last month's designated contract, which is what the index holds as the month starts.
- **Next contract** ("new", "rolled-into", "deferred"): this month's designated contract.
- If lead = next there is **no roll** that month for that commodity. If they differ, the commodity rolls during the roll window.
- A designated month earlier in the year than the current month means next year's contract. For example, gold's December entry G is February of the following year.

```
lead(c, month m) = designated(c, m-1)      next(c, month m) = designated(c, m)
```

3.2 BCOM schedule for every commodity

The shaded cells are the months in which that commodity rolls. “+1” means a contract in the following calendar year. Notice how different the rows are, even within one sector.

Root	Commodity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
CL	WTI Crude	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
CO	Brent Crude	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
NG	Natural Gas	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
HO	ULS Diesel	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
RB	RBOB Gasoline	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
GC	Gold	J	J	M	M	Q	Q	Z	Z	Z	Z	G+1	G+1
SI	Silver	K	K	N	N	U	U	Z	Z	Z	H+1	H+1	H+1
HG	Copper	K	K	N	N	U	U	Z	Z	Z	H+1	H+1	H+1
LA	Aluminum	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
LN	Nickel	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
LX	Zinc	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
ZC	Corn	H	K	K	N	N	U	Z	Z	Z	H+1	H+1	H+1
ZS	Soybeans	H	K	K	N	N	X	X	X	X	F+1	F+1	H+1
ZW	Wheat	H	K	K	N	N	U	Z	Z	Z	H+1	H+1	H+1
KE	HRW Wheat	H	K	K	N	N	U	Z	Z	Z	H+1	H+1	H+1
ZM	Soybean Meal	H	K	K	N	N	Z	Z	Z	Z	F+1	F+1	H+1
ZL	Soybean Oil	H	K	K	N	N	Z	Z	Z	Z	F+1	F+1	H+1
KC	Coffee	H	K	K	N	N	U	Z	Z	Z	H+1	H+1	H+1
SB	Sugar	H	K	K	N	N	V	V	H+1	H+1	H+1	H+1	H+1
CT	Cotton	H	K	K	N	N	Z	Z	Z	Z	H+1	H+1	H+1
LC	Live Cattle	J	J	M	M	Q	Q	V	V	Z	Z	G+1	G+1
LH	Lean Hogs	J	J	M	M	N	Q	V	V	Z	Z	G+1	G+1

Table 1. Designated (post-roll) contract by calendar month, BCOM-style illustrative schedule. Orange cells mark roll months.

3.3 Why each commodity's schedule looks different

The schedules are not arbitrary. Four forces shape every row:

- **Which months are listed and liquid.** The index can only hold contracts that exist and trade heavily. Corn's benchmark months are H K N U Z. Gold's liquidity sits in even months. Crude lists every month.
- **Getting out before delivery.** The index must exit each contract before its first notice or last trading day, so the roll is placed in the month or two before that.
- **Turnover versus staying near the front.** BCOM holds odd-month energy contracts and rolls every *two* months, which halves trading. The S&P GSCI rolls energy every month to stay in the most liquid, nearest contract.

- **Crop seasons.** Grain schedules jump over the old-crop/new-crop switch. Soybeans go from July (N) straight to November (X), skipping thin August and September. Corn and wheat hold December (Z) through the harvest.

Root	Rolls in	Pattern	Why
CL	Feb, Apr, Jun, Aug, Oct, Dec	H K K N N U U X X F F H	All months listed. BCOM holds odd months only, so it rolls every 2nd month and trades less.
NG	Feb, Apr, Jun, Aug, Oct, Dec	H K K N N U U X X F F H	Same as crude. The curve is seasonal (winter premium), so the roll month drives roll yield.
GC	Jan, Mar, May, Jul, Nov	J J M M Q Q Z Z Z Z G G	Liquidity is in G J M Q Z. Rolls in odd months; skips thin October and holds December.
SI	Jan, Mar, May, Jul, Oct	K K N N U U Z Z Z H H H	COMEX H K N U Z. Holds December across the autumn and rolls into March in October.
HG	Jan, Mar, May, Jul, Oct	K K N N U U Z Z Z H H H	Like silver: follows the most active COMEX months.
ZC	Feb, Apr, Jun, Jul, Oct	H K K N N U Z Z Z H H H	Crop year. Holds December (new crop) through harvest, then March.
ZS	Feb, Apr, Jun, Oct, Dec	H K K N N X X X F F H	Jumps from N (July) to X (November, new crop) to avoid thin Q and U.
SB	Feb, Apr, Jun, Aug	H K K N N V V H H H H H	ICE listing H K N V. Holds October only briefly, then March.
LC	Jan, Mar, May, Jul, Sep, Nov	J J M M Q Q V V Z Z G G	Even-month contracts G J M Q V Z. Rolls in odd months.
LH	Jan, Mar, May, Jun, Jul, Sep, Nov	J J M M N Q V V Z Z G G	Adds N (July) in summer, so it rolls in both June and July.

Table 2. Roll months and the reasoning behind each pattern (illustrative).

3.4 Same commodity, different index: BCOM vs S&P GSCI

The same commodity can follow different schedules in different indices. Energy shows this most clearly. BCOM rolls WTI six times a year, while the S&P GSCI rolls it twelve times, always into the next month's contract.

Index	Root	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
BCOM	CL	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
BCOM	NG	H	K	K	N	N	U	U	X	X	F+1	F+1	H+1
S&P GSCI	CL	H	J	K	M	N	Q	U	V	X	Z	F+1	G+1
S&P GSCI	NG	H	J	K	M	N	Q	U	V	X	Z	F+1	G+1

Table 3. WTI and natural gas designated contracts in each index.

Because the roll months differ, **the two indices can hold different WTI contracts on the same day.** In January, BCOM holds March (H) all month and does not roll. The S&P GSCI rolls from February (G) to March (H) on business days 5–9. If your desk tracks both, their crude exposure will be on different points of the curve for parts of every month.

Roll calendar: which BCOM commodities roll in each month

Month	#	Commodities rolling (BCOM)
Jan	5	GC, SI, HG, LC, LH
Feb	17	CL, CO, NG, HO, RB, LA, LN, LX, ZC, ZS, ZW, KE, ZM, ZL, KC, SB, CT
Mar	5	GC, SI, HG, LC, LH
Apr	17	CL, CO, NG, HO, RB, LA, LN, LX, ZC, ZS, ZW, KE, ZM, ZL, KC, SB, CT
May	5	GC, SI, HG, LC, LH
Jun	18	CL, CO, NG, HO, RB, LA, LN, LX, ZC, ZS, ZW, KE, ZM, ZL, KC, SB, CT, LH
Jul	9	GC, SI, HG, ZC, ZW, KE, KC, LC, LH
Aug	9	CL, CO, NG, HO, RB, LA, LN, LX, SB
Sep	2	LC, LH
Oct	18	CL, CO, NG, HO, RB, SI, HG, LA, LN, LX, ZC, ZS, ZW, KE, ZM, ZL, KC, CT
Nov	3	GC, LC, LH
Dec	11	CL, CO, NG, HO, RB, LA, LN, LX, ZS, ZM, ZL

Table 4. Even months are busy: most energy, industrial metals and grains roll together. In odd months only the precious metals and livestock roll.

3.5 The roll window: part in the old contract, part in the new

This is the pattern you see at work: **for a few days the same commodity sits partly in the old contract and partly in the new one.** The index does not switch contracts in one go. It moves a fixed fraction of the quantity at each close across a window of business days:

- **BCOM:** business days 6, 7, 8, 9 and 10 of the month. 20% of the quantity moves at each close.
- **S&P GSCI:** business days 5, 6, 7, 8 and 9. 20% moves at each close.
- Business days are index business days, meaning days the index is calculated, not calendar days.

```
f(t) = min(max(bday(t) - start + 1, 0), 5) / 5
      = share of the quantity already in NEXT after day t's close
q_lead(t) = (1 - f(t)) · Q      q_next(t) = f(t) · Q
```

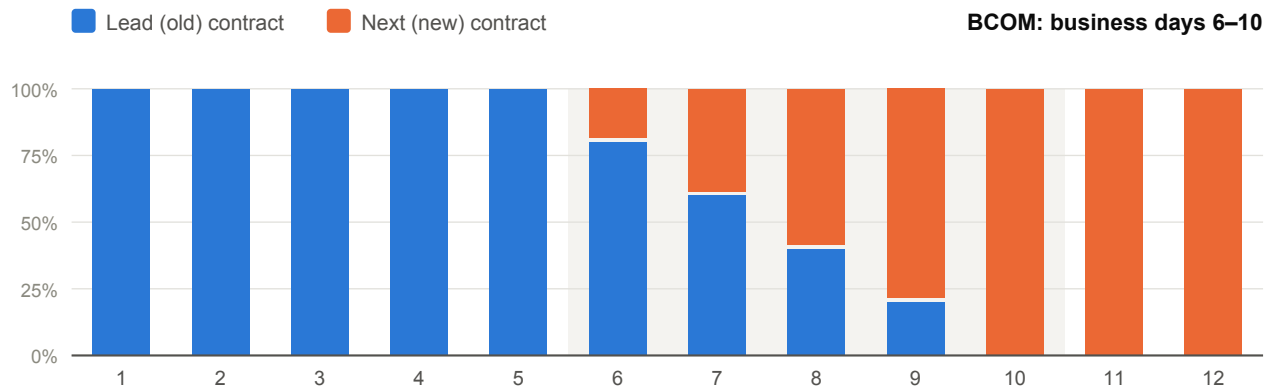


Figure 2. BCOM share of one commodity's quantity held in each contract at the close of each business day of a roll month. The grey band is the roll window.

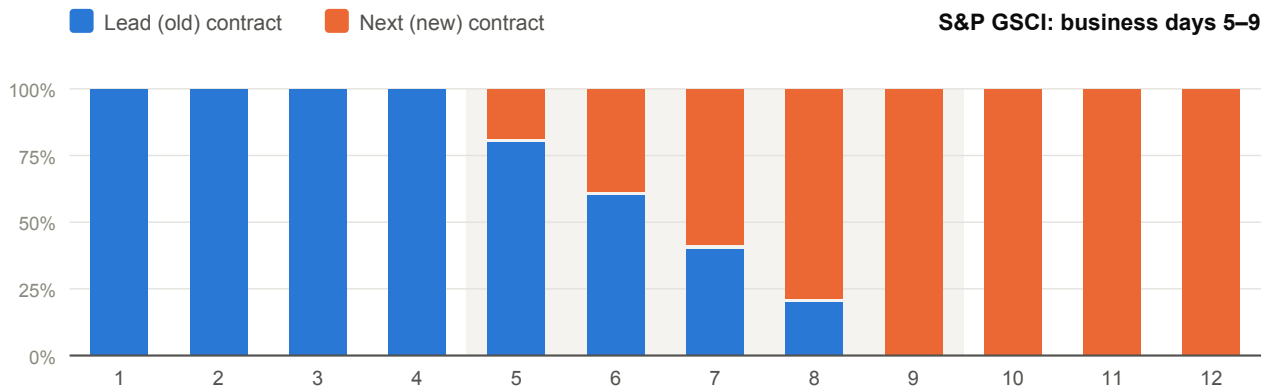


Figure 3. The S&P GSCI window starts one business day earlier.

Business day of month	1–4	5	6	7	8	9	10	11+
BCOM: lead / next	100 / 0	100 / 0	80 / 20	60 / 40	40 / 60	20 / 80	0 / 100	0 / 100
GSCI: lead / next	100 / 0	80 / 20	60 / 40	40 / 60	20 / 80	0 / 100	0 / 100	0 / 100

Table 5. Percent of the commodity's quantity in the lead / next contract after each close.

Why spread the roll over five days? Billions of dollars track these indices. Rolling all of it at one settlement would move prices against the index, and other traders could trade ahead of a known date (the "Goldman roll"). Spreading the roll over five settlements reduces that impact.

Why at settlement? Index trackers typically execute with Trade-at-Settlement (TAS) orders, so they get exactly the price the index uses and do not add tracking error.

3.6 Worked example: BCOM WTI roll, February 2022

Date	BD	f	Lead	q lead	P lead	Next	q next	P next	Value in next	CL contrib.
07-Feb	5	0.0	CLH22	0.1433	76.28	CLK22	0.0000	76.62	0.0%	+0.269%
08-Feb	6	0.2	CLH22	0.1147	74.18	CLK22	0.0287	74.50	20.1%	-0.291%
09-Feb	7	0.4	CLH22	0.0860	75.72	CLK22	0.0573	76.01	40.1%	+0.215%
10-Feb	8	0.6	CLH22	0.0573	77.35	CLK22	0.0860	77.67	60.1%	+0.229%
11-Feb	9	0.8	CLH22	0.0287	77.05	CLK22	0.1147	77.44	80.1%	-0.035%
14-Feb	10	1.0	CLH22	0.0000	76.93	CLK22	0.1433	77.27	100.0%	-0.022%
15-Feb	11	1.0	CLH22	0.0000	76.45	CLK22	0.1433	76.80	100.0%	-0.065%

Table 6. Engine output. q = barrels per index point, P = mock settlement prices. "Value in next" is the dollar share of WTI exposure sitting in the May contract.

Read it row by row. Before the window (business day 5) all 0.1433 barrels are in CLH22. At the close of business day 6 the index sells 20% of that quantity of CLH22 and buys the same quantity of CLK22. At the close of day 10 the move is complete. In between, the **quantity** split is exactly 80/20, 60/40, 40/60 and 20/80. The **value** split differs slightly because the two contracts have different prices. Here May trades above March (contango), so the new contract's value share runs a little ahead of its quantity share.

3.7 The daily return on a roll day

Take business day 7 (the second roll day). Yesterday's close left 80% in the lead and 20% in the next contract. Today's return is earned on *that* mix. Today's extra 20% roll happens at the close and counts from tomorrow:

```
CL part of CDR(7) = [ qL(6) · (PL(7) - PL(6)) + qN(6) · (PN(7) - PN(6)) ] / V(6)
= [ 0.1147 · (75.718 - 74.178) + 0.0287 · (76.009 - 74.499) ] / V(6)
= +0.2146%
qL, PL = lead (CLK22) quantity and price; qN, PN = next (CLK22)
V(6) = value of the WHOLE basket at day 6's close (all 22 commodities)
```

Each of the other rolling commodities is handled the same way. Their contributions add up to CDR, and $ER(t) = ER(t-1) \cdot (1 + CDR)$.

3.8 Quantity vs value: why ER never jumps at a roll

In the index basket the roll swaps *quantity for quantity*. The same number of barrels move from March to May. In contango May is dearer, so the basket's dollar value steps **up** at each roll close. That step is not a return. Nobody made money, and the ER formula ignores it because CDR compares the same holdings on two consecutive days.

- The **Spot** index is the basket value itself, so it *does* include the step. It keeps tracking the price level of whatever contract is held.
- The **ER** chains returns, which is equivalent to an investor rolling *value for value*: sell \$X of March, buy \$X of May, and end up with fewer barrels in contango.
- Over time the difference $ER - \text{Spot}$ is the accumulated roll yield: negative in contango, positive in backwardation.

Reconciling to a real position. A tracker holding \$1bn of BCOM does not hold "0.1433 barrels". It holds $(AUM / \text{basket value}) \times q$ of each contract. After every roll day that scaling factor is recomputed, so the dollar amount sold of the old contract equals the dollar amount bought of the new one. The quantity per index point stays at the CIM.

3.9 Variations you will meet

- **Forward / deferred indices** (for example "1-month forward" or "3-month forward" versions). They use the same weights but shift the schedule so the roll happens one or more months earlier, holding contracts further out the curve. This reduces contango drag and avoids the crowded standard roll dates.
- **Enhanced and roll-select variants** choose the contract to hold by rules such as curve shape or liquidity instead of a fixed table.
- **Custom indices and swaps** at banks often copy BCOM or GSCI but change the window: a different start day, 3 or 10 days instead of 5, or uneven daily fractions. If the percentages in your system are not 20% steps, check the term sheet first.

4. Weights: how much of each commodity

4.1 BCOM: target percentages, then multipliers

BCOM starts from **2/3 liquidity** (futures trading volume) plus **1/3 production** (world production value), both usually 5-year averages. It then applies diversification limits:

- No sector (energy, grains, ...) above **33%**.
- No single commodity above **15%**.
- No commodity plus its derivatives above **25%** (for example WTI + Brent + diesel + gasoline, or soybeans + meal + oil).
- Every included commodity at least **2%**.

Sector	Raw (liquidity/production)	After caps (CIP)
Energy	43.0%	32.5%
Grains	19.0%	22.0%
Industrial	12.5%	14.9%
Livestock	5.0%	5.6%
Precious	16.2%	18.8%
Softs	4.2%	6.2%

Table 7. Mock BCOM weights by sector. Energy is capped from 43.0% to 32.5%, and the excess is spread across the other sectors.

The target percentages (CIPs) are turned into **Commodity Index Multipliers** (CIMs), which are quantities, once a year on the determination date:

```
CIM(i) = CIP(i) · K / P(i, next contract, determination date)
K = scaling constant chosen so the new basket has the same value as the old one
```

From that day until the next rebalance the CIMs are **fixed quantities**. The percentage weights then drift with prices: $w(i,t) = \text{CIM}(i) \cdot P(i,t) / \sum \text{CIM} \cdot P$. A commodity that rallies becomes a bigger part of the index until the next January reset.

4.2 S&P GSCI: production quantities

The S&P GSCI weights by **world production** alone. Its Contract Production Weights (CPWs) are quantities, for example barrels of annual world output, adjusted for liquidity. There are no sector caps. The index is therefore heavily energy-weighted: in this mock 61.2% of the value sits in energy on day one. Because it holds production quantities, not target percentages, **a rally in oil makes the GSCI even more oil-heavy**. Nothing pulls the weight back except the next production update.

```
Dollar weight(i,t) = CPW(i) · P(i,t)          TDW(t) = ∑ CPW(i) · P(i,t)
Spot index = TDW(t) / normalising constant
```

5. Annual rebalance in detail

5.1 The annual cycle

When (typical)	BCOM	S&P GSCI
Summer–autumn	Sponsor collects 5-year liquidity and production data	Committee reviews production data (WPQ) and liquidity
Late Oct–Nov	New target weights (CIPs) announced	New CPWs and eligible contracts announced
Early January (determination date)	CIMs fixed using the CIPs and that day's prices (4th business day in this model)	CPWs already set; the normalising constant keeps the spot level continuous
January roll window	Old CIMs are phased out and new CIMs phased in, 20% a day on business days 6–10	Old CPWs phased out and new CPWs phased in on business days 5–9
Rest of the year	Quantities fixed; % weights drift with prices	Quantities fixed; % weights drift with prices

Table 8. The yearly weighting cycle (typical timing; check the sponsor announcements each year).

5.2 Determination: new quantities from new targets

On the determination date the engine converts each CIP into a quantity at that day's price of the contract being rolled into, then rescales the whole basket so that its value equals the old basket. This scaling is what lets the index move from old to new quantities over five days without any jump.

Root	Old CIM (Dec)	New CIM (Jan)	Change	Weight before reset	Weight after reset
GC	0.00617	0.00610	-1.1%	14.6%	15.5%
CL	0.14332	0.13642	-4.8%	11.9%	10.8%
CO	0.11613	0.13993	+20.5%	7.8%	9.1%
NG	2.40437	2.94411	+22.4%	6.0%	6.8%
HG	1.59060	2.25018	+41.5%	5.0%	6.9%
ZC	1.34943	1.16845	-13.4%	7.1%	6.2%
SI	0.13020	0.10570	-18.8%	4.6%	3.7%

Table 9. BCOM rebalance completed 13 Jan 2023 (engine output). Commodities that fell during the year (copper, natural gas) get more contracts. Those that rose get fewer. Each is pulled back toward its target %. “After” is measured at the end of the window, so it already includes a few days of price moves since the determination date.

5.3 Implementation: rebalance and roll in the same window

The new quantities are not traded in one go. They are phased in during the January roll window with the same 20%-a-day pattern. The general formula covers both the roll and the rebalance at once:

$$q_{\text{lead}}(t) = (1 - f(t)) \cdot Q_{\text{old}} \quad q_{\text{next}}(t) = f(t) \cdot Q_{\text{new}}$$

That produces two cases you will see in position reports:

Case A: the commodity also rolls in January (gold, G → J)

The old quantity of the February contract (G) is sold down while the *new* quantity of the April contract (J) is built up. Both the contract and the quantity change in the same trade.

Date	BD	f	GCG23 (old qty)	GCJ23 (new qty)	Total GC qty	CLH23 qty (Case B)
05-Jan	4	0.0	0.006172	0.000000	0.006172	0.14332
06-Jan	5	0.0	0.006172	0.000000	0.006172	0.14332
09-Jan	6	0.2	0.004938	0.001221	0.006158	0.14194
10-Jan	7	0.4	0.003703	0.002441	0.006145	0.14056
11-Jan	8	0.6	0.002469	0.003662	0.006131	0.13918
12-Jan	9	0.8	0.001234	0.004883	0.006117	0.13780
13-Jan	10	1.0	0.000000	0.006104	0.006104	0.13642
16-Jan	11	1.0	0.000000	0.006104	0.006104	0.13642

Table 10. January 2023 rebalance window from the engine. Gold moves contract and quantity. WTI stays in March (H) and only its quantity changes.

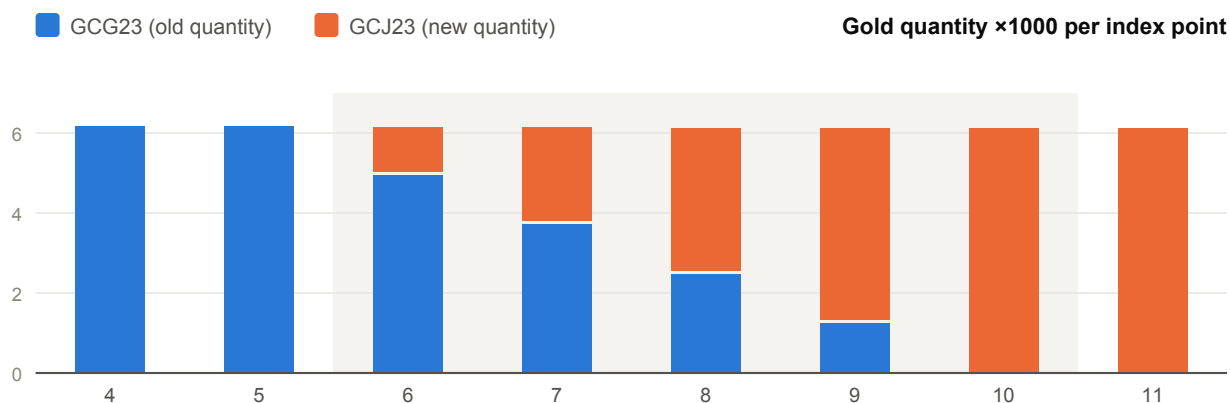


Figure 4. Gold during the January window. The old contract is sold down while the new one is built up at the new CIM.

Case B: the commodity does not roll in January (BCOM WTI stays in H)

Lead and next are the same contract, so the formula collapses to $q(t) = (1-f) \cdot Q_{\text{old}} + f \cdot Q_{\text{new}}$ on *one* contract. The quantity walks in five equal steps from 0.14332 to 0.13642. On a position report this looks like a small daily trade in the same contract, not a roll.

5.4 What actually gets traded

Root	Weight before	Target	Trade on \$10bn	Direction
GC	14.6%	15.0%	\$39m	buy
CL	11.9%	10.8%	\$109m	sell
CO	7.8%	9.3%	\$156m	buy
NG	6.0%	7.5%	\$150m	buy
HG	5.0%	6.9%	\$190m	buy
ZC	7.1%	6.2%	\$94m	sell

Table 11. Approximate rebalance flow for \$10bn tracking BCOM (weight change × AUM). This is on top of the normal roll turnover, which is why January windows are so heavily traded.

5.5 How the S&P GSCI rebalance differs

- The GSCI does **not** reset to target percentages. New CPWs reflect updated production statistics, so year-to-year changes are usually small.
- Weight drift during the year is therefore mostly *kept*. An energy rally stays in the weights, while BCOM would cut energy back each January.
- Mechanically the phase-in looks the same: old CPW on the lead contract, new CPW on the next contract, 20% a day on business days 5–9.

6. Market disruption events

The roll and the rebalance both assume the index can trade every contract at settlement on each window day. A **Market Disruption Event** (MDE) is a day when that assumption fails for a specific contract. The rules protect the index, and the people replicating it, from being forced to "trade" at a price nobody could actually trade at.

6.1 What counts as a disruption

Event	Example	Why it blocks trading
Limit price at settlement	Live cattle settles limit-down after a disease headline	At limit the market is one-sided; nobody can sell at that price
Trading suspended or terminated	Exchange halts a contract, or the LME suspends nickel	No market at all
Settlement price not published	Exchange system outage	No price to trade or value at
Not a trading day for that contract	Exchange holiday on an index business day (for example Good Friday on some venues)	The exchange is closed

Table 12. Typical disruption triggers. The exact list and wording are in each index's rulebook.

6.2 Disruption affects rolling, not valuation

An important distinction: on a disrupted day the index is still **valued**, using the published settlement price (the limit price if limit-locked, or the last available price if none was published). What stops is **trading**, meaning the roll or rebalance fraction scheduled for that day.

- The disruption applies **per commodity** (per contract). Every other commodity rolls normally.
- The portion that was due is **postponed** to the next business day with no disruption for that commodity, and it is rolled on top of that day's own scheduled portion.
- The disrupted commodity can end up rolling after the normal window has closed, so its roll finishes later than everyone else's.
- If the disruption lasts many days, the rulebook sets a cut-off after which the sponsor or calculation agent completes the roll using its judgement of a fair price. Check the exact number of days in the current methodology.

6.3 Worked examples (BCOM window, days 6–10)

Business day	5	6	7	8	9	10	11	12	13
Normal	0%	20%	40%	60%	80%	100%	100%	100%	100%
A: MDE on days 7–8	0%	20%	20%	20%	80%	100%	100%	100%	100%
B: MDE on day 10 (last)	0%	20%	40%	60%	80%	80%	100%	100%	100%
C: MDE on days 9–11	0%	20%	40%	60%	60%	60%	60%	100%	100%

Table 13. Share of the disrupted commodity in the NEXT contract after each close. Red cells are disrupted days: nothing moves, and the missed portion is rolled on the next clean day.

Case A. Live cattle settles limit-down on days 7 and 8. Its roll freezes at 20% after day 6. On day 9 it rolls the 40% it missed plus day 9's own 20%, jumping from 20% to 80%. Day 10 finishes normally, so the window end is unchanged but day 9's trade is three times its normal size.

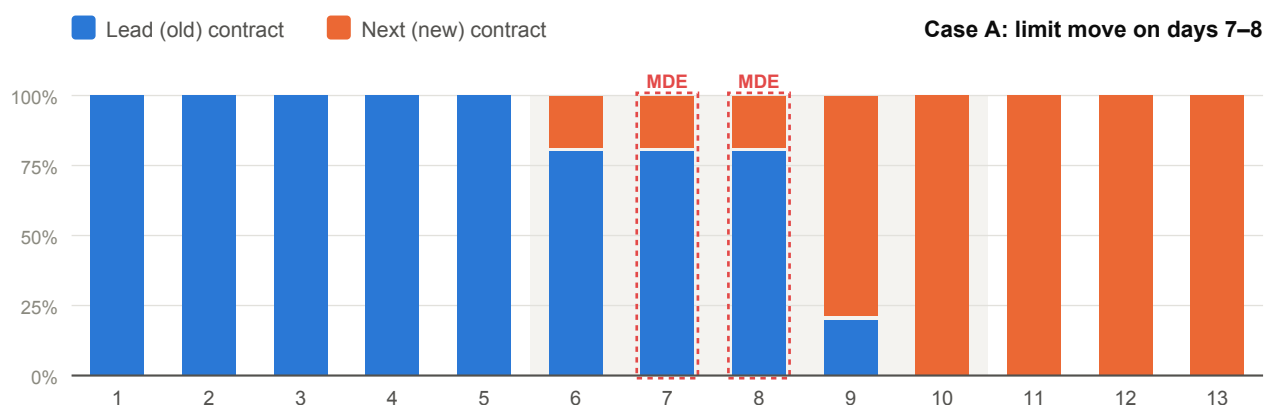


Figure 5. Case A. The dashed red outlines mark disrupted days on which the split is frozen.

Case B. The disruption hits the last scheduled day. The final 20% cannot move, so this commodity rolls it on day 11, **one day after the window has closed for everything else**. On day 10 your position report shows 20% still in the old contract, and this is correct.

Case C. Three disrupted days in a row (9, 10 and 11). The commodity is stuck at 60% until day 12, when it rolls the remaining 40% at once. If a disruption like this ran past the rulebook's cut-off, the sponsor would finish the roll at a price it judges fair.

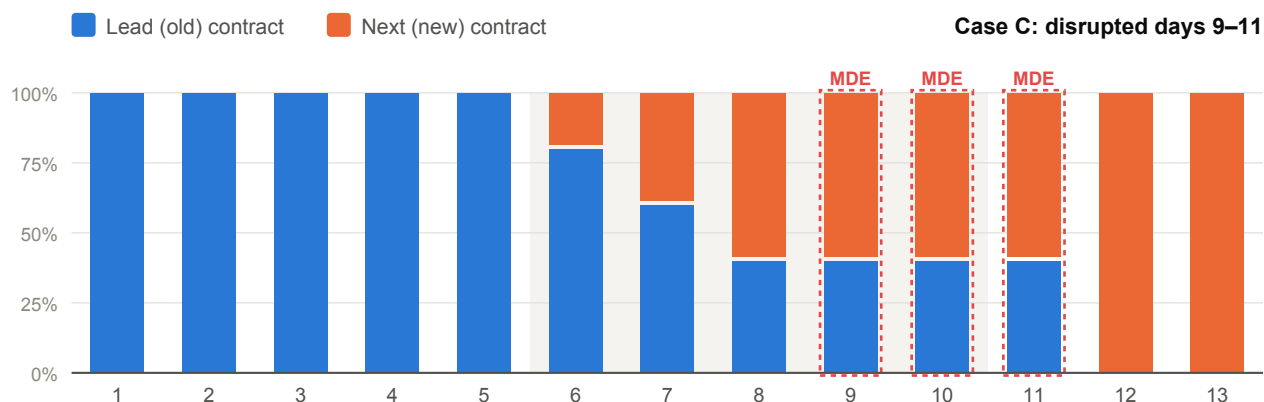


Figure 6. Case C. The roll is completed two business days after the normal window.

6.4 Disruption during the January rebalance

- A rebalance is simply a roll with new quantities, so the same postponement applies. If gold is disrupted on day 8 of January, both gold's contract switch and its CIM change for that day are deferred.
- For a commodity that does not roll in January (Case B in 5.3), a disruption postpones only its quantity step on that day.
- If a commodity is disrupted on the **determination date**, the rulebook specifies which price is used to set its new CIM, typically the next non-disrupted settlement. The CIMs are still fixed before the window opens.

6.5 Holidays

Index business days follow the index's own calendar, but each exchange has its own holidays. When an index business day is a holiday for one exchange (for example LME closed on a UK bank holiday while CME trades), that commodity is treated as disrupted for rolling purposes. Its return that day is zero, because its price is carried forward, and its roll portion moves to the next open day. This is the most common reason one commodity's roll lags the rest by a day.

Checklist: my position's split isn't a clean 20% step. Why?

1. Is the commodity disrupted today or on an earlier window day (limit move, halt, missing settlement)?
2. Is its exchange closed today (a holiday that is not an index holiday)?
3. Is it January, so rebalance quantities are changing as well as contracts?
4. Is the product a variant (forward, enhanced, custom window) rather than the standard index?
5. Are you comparing quantity shares with value shares? In contango they differ during the window (Section 3.6).

7. Putting it together: ER, TR and Spot



Figure 7. Mock BCOM levels, base 100 on 2022-01-03. The gap between Spot and ER is roll yield; the gap between ER and TR is T-bill interest.

Index	Spot	Roll yield	= ER	Collateral	= TR
BCOM	-8.02%	-3.88%	-11.90%	4.33%	-7.57%
S&P GSCI	-11.76%	-2.25%	-14.01%	4.23%	-9.78%

Table 14. Annualised decomposition on mock data 2022-01-03 to 2025-12-31 (approximate additive split).

7.1 Daily calculation checklist

1. Load settlement prices for every held and next contract, and the 91-day T-bill rate.
2. Flag disruptions per contract (limit, halt, missing price, exchange holiday).
3. Compute CDR on yesterday's closing holdings. Every contract is valued, even disrupted ones.
4. Update ER and TR (and Spot from today's closing holdings).
5. If it is the determination date, fix new CIMs/CPWs, scaled for value neutrality.
6. If today is inside the roll window, move each commodity to its target split. Disrupted commodities stay put and catch up on the next clean day.
7. Store today's closing holdings. They drive tomorrow's CDR.

7.2 Glossary

Term	Meaning
CIP	Commodity Index Percentage: BCOM target weight after caps
CIM	Commodity Index Multiplier: BCOM quantity per index point, fixed for a year
CPW	Contract Production Weight: S&P GSCI quantity, based on world production
CDR	Contract Daily Return: the one-day return of the futures basket

Term	Meaning
TBR	Treasury Bill Return used for total-return indices
Designated contract	The contract the index holds after a month's roll
Lead / next	Contract rolled out of / into during the current month
Roll window	Business days over which the roll is spread (BCOM 6–10, GSCI 5–9)
Determination date	Day new weights are turned into quantities
MDE	Market Disruption Event: a day a contract cannot be traded at a fair settlement
Roll yield	Return from moving along the curve: negative in contango, positive in backwardation
TAS	Trade-at-Settlement: order type that fills at the day's settlement price

7.3 Where each concept lives in the code

Concept	File	What to look at
Designated contracts	indexpricing/contracts.py	BCOM_SCHEDULES, GSCI_SCHEDULES, designated_contract()
Roll window and split	indexpricing/engine.py	roll_fraction(), lead_next(), positions()
ER / TR / Spot	indexpricing/engine.py	RollingFuturesIndex.run()
Rebalance phase-in	indexpricing/engine.py	pending quantities, value-neutral scaling
BCOM caps and CIMs	indexpricing/weights.py, indices.py	bcom_target_weights(), build_bcom()
GSCI CPWs	indexpricing/weights.py, indices.py	gsci_production_quantities(), build_gsci()
Disruption deferral	not yet in the engine	Section 6 logic (docs/build_guide.py: disrupted_path)